Greenhouse Games

Public release — design document v0.2

Project	Greenhouse Games — public release
Owner	Helayna Bostian, HBGreenhouses LLC
Status	Design complete; ready for Phase 1 vertical slice
Version	v0.2 — supersedes v0.1
Last updated	2026-10-22

Changes since v0.1

• **Art direction flipped** from GTA-grade realism to stylized 2D illustration with a single art bible. Rationale in §7.

• **Progression system added** (§6.7). Day-over-day business growth, unlockable greenhouse types, recurring clients, regional reputation, 90-day season arc.

• **Time-of-day and weather as mechanics** (§6.8), not just UI readouts.

• **Localization-ready strings from day 1** (§7.5). Every UI string flows through t(key); translation deferred, infrastructure not.

• **Tooling added** (§8): content verify script, runtime smoke test, state-jump debugger, tuning panel. Build order updated.

• **Intro cutscene** (§5.0) added — 60s built from in-game scenes.

• **Module layout defined** (§8.2) so parallel coding sessions don't collide.

• **Cut: multiplayer, 3D (Three.js) consideration.** Never on our list, now explicit.

• **Public-safety expanded** (§7.4) to cover real-client resemblance in "slimy" archetypes.

1. What this is

A single-player simulation game about running a greenhouse installation business. The player is "you" — the owner-operator of a small greenhouse company — and the game plays out as a day-in-the-life of that work: triaging morning leads, driving to sites, scoping clients, writing quotes, running installs, handling the slimy, the grimy, and the good. Phone-first fiction (HBGreenhouses OS) with illustrated scene cards for the world outside the phone.

Working title

Greenhouse Games

Platform target

Web first (hbgreenhouses.com), native iOS and Android second via Capacitor wrapper.

Rating target

12+. Occasional mild language, no violence, no sexual content.

2. What this is *not*

- Not the training game. Training game stays internal with its real trade data.
- Not a reveal of HBGreenhouses trade specifics. No real FRAMEPARTS, GASKETS, CUTS, SITES. Scoping philosophy and client-education are fair game; manufacturing process is not.
- Not photoreal. Stylized 2D illustration — intentional, consistent, warm. See §7.
- Not multiplayer. Not 3D. Not Three.js.

3. Core loop

One day of the game:

1. **Lock screen** — wake, triage overnight notifications
2. **Home screen** — HBGreenhouses OS, 8 apps
3. **Morning triage** — Phone, Messages, Mail, Weather
4. **Commit** — Calendar books the day, Maps shows the route
5. **Drive** — stylized top-down map + illustrated event cards
6. **Site visit OR Install day**
7. **Quote write-up + client response** OR **Walk-through + invoice**
8. **End of day** — wrap, stat shifts, next-day preview
9. **Next lock screen** — new day, threads carry forward

~15 min per in-game day. v1 ships with 90 days of content organized as three 30-day chapters. First 7 days scaffold complexity like a tutorial without being one.

4. HBGreenhouses OS

Fictional phone operating system, branded HBGreenhouses. Reads like iOS to the player; has its own identity, icons, and voice. Dark mode default.

4.1 Brand standards (locked)

- **Colors:** Dark #0a0a0a · HB Green #22863a · Accent Green #4ade80 · Light BG #f5f5f0 · White · Grey #666666
- **Fonts:** Inter (UI/body) · Permanent Marker (logo/display)
- **Logo:** Mom_Logo.png. Transparent PNG + SVG required from Trae for the repo.
- **Design bar:** Apple-grade cinematic polish within the stylized 2D direction. Benchmark: fangieshop.com

4.2 The leaf motif

Leaf from the logo lifted as a system accent. Appears in Accent Green on the status bar of every screen. Tiny signature, not full logo.

4.3 Apps (v1 home screen)

App	Role
Phone	Call log, voicemails. Spam filter is an upgrade.
Messages	Crew, clients, Steph. Fast decisions. Steph has instinct level.
Mail	Longer leads, quotes, referrals.
Weather	Install window forecast. The clock the game runs on.
Maps	Routes, drive time, traffic. Reputation regions light up as earned.
Camera	Site photos. Scoping and install documentation.
Reviews	Client reviews. Currency for upgrades.
Bank	Balance, deposits, outgoing.

Deferred for v1: Calendar (inline), Notes, About, Settings.

4.4 OS fingerprint

Every app has at least one Accent-Green left-bar callout — the system's house voice. Examples:

- Phone: "Unlocked by listening"
- Messages: "Steph's instinct · lv 2"

- Mail: "Scoping cues"
- Weather: "Install window"
- Quote: "Close read · likely outcome"
- End of day: "What today taught you"

5. Scene inventory

Every scene with its purpose and reuse pattern.

5.0 Intro cutscene (NEW)

60 seconds. Built from in-game scenes with text overlays. Sets context: you just took over the business, phone's ringing, here's your week. Skippable after first viewing.

5.1 Lock screen

Opens every day. Date, time, weather summary, stacked notifications. Swipe up to unlock.

5.2 Home screen

HBGreenhouses OS, 8 apps in 4-column grid, weather + install-window header, dock: Phone/Messages/Mail/Camera.

5.3 Phone — Recents / Voicemail detail

Call log with spam filter visible. Voicemails have mangled transcripts; tapping unlocks location + urgency + scope read. Spam filter is a leveled upgrade.

5.4 Phone — Call in progress

Live notes panel surfaces location pins, scope reads, decision points *as the call happens*. Timer, mute, speaker, end.

5.5 Messages — thread

iMessage-style. Lowercase texting voice. Steph's instinct callout appears when she flags something.

5.6 Mail — inbox and detail

Color-coded tags: REFERRAL, LOW BUDGET, UNUSUAL, STEPH VERIFIED. Detail surfaces Scoping cues. Primary action: Book site visit.

5.7 Weather

Current conditions, 7-day install-window bar (green/yellow/red), forecast list, wind + ground tiles with installer-relevant readings.

5.8 Calendar

Month mini-view, today's timeline, bookings from Mail land here. HOLD items for maybes. Day-read callout.

5.9 Maps

Stylized hand-drawn route. Segments colored by condition. Floating ETA card. Pre-drive tiles: Fuel, Trailer. Start drive CTA. Regions light up with reputation as you earn jobs in them.

5.10 Drive — base top-down

Stylized map (NOT photoreal). Truck crawling route. Speed/fuel/distance HUD. Trail. Event markers along the way. Phone notifications pull down mid-drive.

5.11 Drive — event card

Illustrated card scene in the art bible style. One line of game voice + 3 choices with transparent costs + 10s timer with default. Reusable.

5.12 Drive — skill section (rare)

Pure driving mechanic for terrain that earns it. Not every drive.

5.13 Site visit — scoping cards

Pick 4 of 6: Sun, Drainage, Power+Water, Access, HOA/Permits, What they grow. Opens with client greeting over illustrated scene.

5.14 Site visit — site read

Post-visit summary. Green/yellow/red cards for asked questions, grey for skipped. Scope-read callout. Write up the quote CTA.

5.15 Quote write-up

Scope summary auto-filled. Price slider with market range + fair/lowball/greedy bar. Three honesty-flag toggles. Close-read predicts outcome.

5.16 Client response (email reply)

Fast-reply tag = positive signal. Deal-signal callout. 3-choice response.

5.17 Install day — load out

Pack list grid. Required items pre-checked with why. Optional slots dashed. Weight warning at threshold. Scope-read rewards good listening.

5.18 Install day — on-site stages

4-stage tracker. Setbacks appear as red cards with 3 honest choices. Progress bar. Crew footer with energy.

5.19 Install day — walk-through

Client asks a question. 3 answer choices: modest, honest, educational. Education-moment callout for teaching. Send invoice CTA.

5.20 End of day

Tone-calibrated opening line. Jobs list, stat tiles (Bank/Reviews/Karma/Energy), first-review card if earned, What today taught you (3 extracted lessons), Tomorrow preview.

5.21 Next-day lock screen

Pattern repeats. Day counter visible. Carry-over notifications. Ongoing threads surface.

5.22 Chapter end (NEW)

Day 30, 60, 90. A 90-second recap: season stats, biggest wins, hardest days, next chapter tease. Light narrative beat, not a boss fight.

6. Systems

6.1 Stats

- **Bank** — currency, deposits in, invoices out
- **Reviews** — 1–5 star reviews, averaged, counted
- **Karma** — reputation from choices
- **Energy** — daily resource, depletes with work and decisions

6.2 NPC levels

- **Steph · Instinct** (1–5). Flags become more accurate.
- **Trevor · Skill** (1–5). Faster, cleaner install stages.
- Future: Trae, Hayyah, Giorgia.

6.3 Upgrades (reviews currency)

Cost	Upgrade
3	Spam filter lv 2
5	Second truck (parallel jobs possible)
8	Hire Trevor full-time
10	Weather pro (72-hr wind detail)
15	Steph full-time (instinct unlocks lv 3)

Verify rule: every upgrade must keep the honesty/decision loop intact. If an upgrade removes a tradeoff, cut it.

6.4 Honesty flags (quote mechanic)

Transparent moral weight. Each flag has a real $ cost and a close-rate/trust benefit. Player sees both before toggling. Encodes HBG's client-first philosophy as game logic.

6.5 Setbacks (install mechanic)

Red cards during stages. Three choices: right way (slow, builds trust), hack (fast, hidden future cost), honest retreat (tarp it, come back). Never game-over. Always recoverable.

6.6 Save state

localStorage on web (try/catch, game plays with storage blocked). Native storage on iOS/Android. One JSON blob, < 1 MB, versioned. Export/import for safety.

6.7 Progression (NEW)

The retention loop. v0.1 had scenes; v0.2 has a reason to keep playing.

- **Greenhouse types unlock over time.** Lean-to repairs day 1 → full lean-to builds day 10 → Victorian day 25 → treehouse day 50 → commercial day 75.
- **Reputation regions on Maps.** Start grey. Earn a job in a county, that county tints green. Later jobs there pay more and close faster.
- **Recurring clients.** Patel comes back for maintenance. Winkler refers two more. Lori sends a cousin.
- **Business growth.** Second tech, second truck, second service area — each unlocks parallel jobs.
- **Season arc.** 90 days split into 3 chapters. Chapter 1 (days 1-30): survive and build reputation. Chapter 2 (31-60): grow and specialize. Chapter 3 (61-90): run a real business.
- **Chapter ends** show a recap scene (§5.22) and tease the next chapter.

6.8 Time of day and weather as mechanics (NEW)

- **Time of day** drives UI tinting (dawn, midday, afternoon, dusk, night) via CSS filter over scene art. Cheap, atmospheric, consistent.
- **Weather** changes gameplay: rain slows drives, cancels install days, triggers specific events (tarp emergency, call-off text from a client).
- **Install window** is computed from weather + daylight. The player sees it on Weather and Calendar.
- **Day clock** runs through every scene — a tap-heavy morning can push you into dusk, which affects what's still possible.

7. Content pipeline

7.1 Art direction (CHANGED FROM v0.1)

Stylized 2D illustration, single bible, single visual voice. Not photoreal. Think modern editorial illustration — grounded, warm, intentional. Reference anchors to be chosen by Helayna before Phase 2. Why: the postmortem evidence is that AI-generated photoreal scenes fail at consistency, and realism exposes every flaw. A stylized bible we can hold a line on ages better and ships faster.

7.2 Art bible contents

- Style anchor image(s) — Helayna's pick
- Character sheets — every recurring NPC drawn once from 3 angles
- Prop library — truck, trailer, tools, greenhouse model types
- Palette — tied to brand kit; grounds (grass, soil, concrete) and sky palettes defined
- Light recipes — dawn / midday / dusk / night as CSS filter values
- Weather overlays — rain, cloud, overcast as reusable layers
- Composition rules — camera height, framing, where UI chrome sits

7.3 World art workflow

Every scene card:

1. Scene designed in-chat with placeholder + art brief
2. Brief + bible references sent to illustrator (human or AI-assisted)
3. Art returned, run through §9 consistency checklist
4. Composited into UI as background with time-of-day filter

7.4 Public-safety line (expanded)

- No real FRAMEPARTS, GASKETS, CUTS, SITES. No crescent-28, no butterfly-72. No Royal Victorian build order.
- No real HBGreenhouses client names, likenesses, or situations verbatim. Composite characters only.
- "Slimy" client archetypes must not resemble any identifiable real client — Trae reviews the cast before each content release.
- Crew member names (Trevor, Steph, Trae, Hayyah, Bud, Gregg) appear with their consent. Any crew member who declines gets a composite replacement.
- Scoping philosophy and client-education principles are HBG public brand and encouraged.

7.5 Localization-ready strings (NEW)

Every UI string flows through `t('key.subkey')` reading from `locales/en.json`. English is the only locale shipped in v1. Translation is a flip-the-switch operation if the game earns it. Day-1 cost: discipline. Day-100 retrofit cost: weeks of boring work. Non-negotiable.

7.6 Writing voice guide

- **Game voice** (narration, callouts): calm, warm, specific. One-sentence lessons. Never lectures.
- **Client voice**: varies by character. Patel = polite quality-buyer. Castellanos = low-budget cheerful. Mystery caller = mangled.
- **Steph voice**: lowercase, abbreviated, warm, occasional emoji.
- **System language**: sentence case. Buttons are verbs. Errors are honest.

7.7 Tool licensing (open)

Every AI tool used for assets (image, 3D, audio, font) must have its commercial-use terms verified before assets land in the repo. Jam-tier free plans are not commercial licenses. This is a research task before Phase 2 (asset production), not a Phase 1 blocker.

8. Engineering

8.1 Module layout

Partitioned so parallel coding sessions don't collide.

```
src/ engine/ game loop, time, save/load — rarely touched data/ leads.json, events.json, clients.json — content phone/ home, lock, status bar apps/ phone/, messages/, mail/, weather/, maps/, calendar/, camera/, reviews/, bank/ scenes/ drive/, site-visit/, install/, end-of-day/ ui/ buttons, callouts, cards — shared primitives state/ player, calendar, reviews, upgrades i18n/ t() + locales/en.json art/ references the illustration bundle tools/ verify.mjs content validator (runs on commit) smoke.mjs headless playthrough test tune/ sliders, dev-only jump/ state loader, dev-only assets/ images, audio, fonts locales/ en.json (es.json etc. later)
```

Rule: one PR touches one folder when possible. Content (data/.json) PRs run verify.mjs on commit.*

8.2 Build tooling

- **Dev server:** Vite (fast, zero-config, hot reload)
- **Native wrapper:** Capacitor for iOS/Android (TBD after web performance confirmed)
- **Package manager:** npm (default, no exotic tooling)
- **Hosting:** GitHub Pages for Phase 1 playable URL; hbgreenhouses.com/games for production
- **No bundler config needed** — Vite handles it

8.3 Dev tools (ship separately from game)

Priority	Tool	When to build
1	verify.mjs	Phase 1 — before first content merge
2	smoke.mjs	Phase 1 — before first deploy
3	state-jump debugger	Phase 2 — when days > 3
4	tuning panel	Phase 4 — when events > 20
5	scenario editor	Only if events > 100 and we are drowning

9. Verification plan

9.1 Content verify (tools/verify.mjs)

Runs on every commit touching data/. Must pass before merge.

- Every lead has: id, name, city, tag, close rate, thread with opening message
- Every event has: trigger, exactly 3 choices, each with cost and benefit
- Every job stage has: name, order, at least one setback variant, an exit
- Every setback has 3 responses with resolvable outcomes
- Every quote path can reach a client response (no softlocks)
- Reward balance: total possible reviews ≤ total upgrade cost in planned length
- Trade-data regex: no FRAMEPARTS, GASKETS, CUTS, SITES tokens; no training-game pane dims; no "crescent" / "butterfly" with lengths
- Public-safety regex: no verbatim real client names from a maintained redacted list

9.2 Runtime smoke test (tools/smoke.mjs)

Headless browser plays Day 1 end-to-end on two paths.

- Safe path: voicemail → Patel → site visit → quote → end of day
- Slimy path: lowballer email, ghost the referral, bad quote
- Asserts: no console errors, every scene renders, save writes and reloads, Day 2 opens
- Font-size + contrast audit (inherits from training game's smoke.mjs)
- Asset-reference check: every image/sound referenced by code exists in assets/
- Storage-blocked variant: run again with localStorage disabled; game must still play

9.3 Art consistency checklist

Manual, run before merging any new scene art.

- Does it match the style bible? (reference-image comparison)
- Is the time-of-day filter correct for the scene?
- Are character faces drawn from the character sheet, not improvised?
- Does the HBGreenhouses logo appear only where approved?
- Any element accidentally revealing trade specifics?
- Any resemblance to a real HBG client, employee, or competitor? If yes, redesign or get consent.

10. Build order

Phase 1 — Vertical slice (requires GitHub repo attached)

- Project scaffold (package.json, Vite, folder layout, GitHub repo)
- State module with save/load (localStorage try/catch, blocked fallback)
- Localization harness (t() + empty en.json)
- UI primitives (button, callout, card)
- Phone shell (status bar, home indicator, back nav)
- Character creation screen
- Day 1 lock screen with three notifications
- Home screen (Phone functional only)
- Phone app: Recents + voicemail detail with Unlocked by listening
- data/leads.json with the Jefferson lead
- tools/verify.mjs v1
- tools/smoke.mjs v1
- First deploy to GitHub Pages — playable URL

Phase 2 — Morning triage complete

- Messages app with Steph thread
- Mail app with Patel referral
- Weather app with install window
- Enough state to triage and pick a job

Phase 3 — Drive and site visit

- Maps app with route
- Drive top-down with one event
- Site visit scoping cards + site read
- First placeholder art (bible not yet needed)

Phase 4 — Quote + response + install day

- Quote write-up
- Client response with 3 choices
- Install day: load out, stages, walk-through
- Tuning panel introduced

Phase 5 — End of day + persistence + Day 2

- End of day wrap
- Save state carries to Day 2 lock screen
- Ongoing-threads system (mystery 770 caller)

Phase 6 — Content expansion

- Days 3–7 scaffold complexity
- More leads, more events, Steph/Trevor leveling
- Upgrade shop

Phase 7 — Progression systems

- Greenhouse type unlocks
- Regional reputation on Maps
- Recurring clients
- Business growth (second truck, hire)

Phase 8 — Art bible production

- Helayna picks reference anchors
- Illustrator onboarded (or AI pipeline locked)
- Replace placeholder art with bible-driven scenes
- Time-of-day + weather overlays

Phase 9 — Chapters and arc

- 90-day content build-out
- Chapter-end recap scenes
- Intro cutscene

Phase 10 — Polish, audio, performance

- Sound design
- Load time + asset size budgets
- Device testing

Phase 11 — Native wrappers and App Store

- Capacitor build for iOS
- Capacitor build for Android
- App Store + Play Store submission

11. Working rules (public game)

1. **Save state is on.** localStorage try/catch, game plays with storage blocked.
2. **Setbacks, not failure screens.** Nothing kills the player.
3. **Every rejection teaches the rule.**
4. **Verify every scene has an exit.** No softlocks.
5. **Public-safe or it doesn't ship.** Trade-data regex and real-client regex both green.
6. **Stylized 2D illustration, one bible.** No photoreal mixing.
7. **OS voice is one voice.** House style enforced across every callout.
8. **Localized strings from day 1.** No English hard-coded.
9. **One PR per folder.** Content and engine never mixed.
10. **verify + smoke green before merge.** No exceptions.
11. **Say plainly what's a guess.** Explicit guesses marked in docs and code.

12. Open questions

- GitHub account — Helayna to sign up; username pending
- Illustration reference anchor — Helayna's pick before Phase 8
- Illustrator — human hire, AI pipeline, or Trae?
- Monetization — free, paid upfront ($2.99), free + one-time unlock, something else?
- AI tool licensing audit — before Phase 2 asset production
- Native wrapper feasibility — depends on Phase 2 web performance test
- Crew consent for likeness use — Trae to collect before any crew art ships
- Audio direction — original score, licensed library, or deferred to Phase 10?

End of document v0.2. Supersedes v0.1. Future updates get a version bump and a dated change-note under the title.

Prepared by Claude for Helayna Bostian & Trae Tucker, HBGreenhouses LLC — 2026-10-22.